School of Informatics & IT
Diploma in Cybersecurity and Digital Forensics

AY23/24 Semester

MP Weekly Project Progress Report

Project Title: OT Security Patch Management System	
Student Name: Isaac Tan	**Adm No:** 2100902B
Supervisor Name: Yap Chern Nam	**Week No:** 01

Tasks Completed

- Created work delegation and project management using ClickUp
- Continuous research about ESP32 and experimenting with it
- Attempted to install NuttX into ESP32

Issue/Risk Tracking

NIL

We are unable to start the project right due to insufficient resources

All online guides are outdated/invalid

Meeting minutes with MP supervisor

During our meeting, we discussed the potential plans for the ESP32. We also discussed the need to accomplish some basics first such as installing NuttX on ESP32, OTA patch update of NuttX, and a simple flashing of LED lights from the ESP32. We also further discussed about researching how to update the LED light application through OTA updates.

Weekly Self-Reflection

So far, the project seems manageable after having just done an IoT project from one of my elective subjects and I am still fresh on the IoT concepts. However, we are already facing a few roadblocks due to outdated and invalid online sources to get the project started. We will continue to research and seek help from our supervisor for help and clarification on the project.

Furthermore, I have high trust and confidence for my groupmates that we can all work well and accomplish this project together, and help each other when in need.

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MP Weekly Project Progress Report

Project Title: OT Security Patch Management System	
Student Name: Isaac Tan	**Adm No:** 2100902B
Supervisor Name: Yap Chern Nam	**Week No:** 02

Tasks Completed

- Managed to get a NuttX shell
- Can use ESP-IDF or MCUBoot bootloader binary files from Espressif GitHub to boot into ESP32
  - When using ESP-IDF bootloader, need to include both bootloader and partition table binary files.
  - When using MCUBoot bootloader, only need to include bootloader binary file; the bootloader binary file is pre-built with a partition table.
- We were able to use another ESP32 devkit module called "`esp32s3-devkit:sta_softap`" to enable WiFi and turn the ESP32 into a mini router (access point) which is detectable from a mobile device

Issue/Risk Tracking

- We are unable to boot NuttX into ESP32 from the secondary slot. There are too many dependencies in the "`make menuconfig`" page and we are still unsure of the underlying issue and are still trying to work on it.
- Even after selecting to boot from the secondary slot in the "`make menuconfig`" page, the message states "`Boot source: none`", indicating secondary slot detection failure.

Meeting minutes with MP supervisor

This week, we had 2 meetings. The first meeting, we presented to our supervisor how we managed to get into the NuttX shell. He then complimented on our progress and gave us a TODO list on how to carry on from there such as booting from the secondary slot instead of the primary slot, enabling WiFi on the ESP32, and more sources to read up on for booting up in the secondary slot.

Weekly Self-Reflection

This week was quite hectic as I spent a few days staying up very late at night to keep trying to boot from the secondary slot, but to no avail. This week we only managed to get the NuttX shell, but it is still the bare minimum of this project. There are still many features and implementations to do and getting the NuttX shell was only the beginning. It still feels extremely overwhelming from the plethora of features to implement, but we are still unable get the basics to work, like booting from the secondary slot.

I am trying my best to contribute as much to get all the basics done and working so we can properly delegate the workload for other implementations that our supervisor wants. We may be slow in getting things to work, but at least there is progress being made.

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Project Title: OT Security Patch Management System	
Student Name: Isaac Tan	**Adm No:** 2100902B
Supervisor Name: Yap Chern Nam	**Week No:** 03

Tasks Completed

- Managed to get 2 ESP32 devices to ping to each other, one as an access point connected to a router, and the other as a lone ESP32.
- Still researching about MQTT through ESP32 and how we are going to make it work, since we can already ping both devices.
- Still trying out the ESP32 LED lights example, continuing on from the few hiccups we had last week and still figuring out how it works.

Issue/Risk Tracking

- The same issue persists, that the root cause may be the menuconfig configurations, that there are many dependencies of each other. We are still figuring out most of the things and understanding the concept first of how it works.
- For the LED lights example, the error thrown was a "failed to open /dev/userleds" error, indicating something relating to a plugin error. It is all still very confusing and complicated and we are still trying our best to figure out these issues. This device is still very new in the industry and there are very few reports and articles about it which makes this entire project extremely tough.

Meeting minutes with MP supervisor

During our meeting minutes this week, our supervisor was telling us what to do after getting 2 ESP32 devices to ping to each other, like having the ESP32 access point to be a DHCP server and automatically give out IP addresses. We found out that the DHCP is enabled by default, and we were able to use several external devices to connect and ping to the access point. We also further discussed about the work delegation in our group, since these few weeks we have been tackling the project altogether and helping out everyone's parts.

Weekly Self-Reflection

This week was also quite hectic as I started to panic more about our progress in this project. We are slow, but at least there is progress being made. We are tackling new arising issues every time we do something new, such as the LED lights example or MQTT. We are all encountering our own different issues and are still trying our best to research and try out certain solutions. I am still trying to stay positive about this project despite our slow progress, because I know my team is working hard alongside with me and our efforts will be worth it in the end. Our supervisor is also very helpful with providing us more information and resources about the ESP32 and we are still trying to make this work.

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MP Weekly Project Progress Report

Project Title: OT Security Patch Management System	
Student Name: Isaac Tan	**Adm No:** 2100902B
Supervisor Name: Yap Chern Nam	**Week No:** 04

Tasks Completed

- Started to set up MQTT publisher and broker in their own separate VMs and able to send files over from the publisher to the broker.
- Able to ping from the AP to Google (8.8.8.8) by setting up wlan0 connections through bridged adapter in the VM.

Issue/Risk Tracking

- We are still stuck at the same progress ever since getting the NuttX shell and still trying out many ways to figure out how to get things working in the shell.
- We are still trying to figure out how to configure the ESP32 end device to be an MQTT subscriber. We already enabled MQTT options in menuconfig, but we have yet to figure out how to configure it into a subscriber in the NuttX shell.

Meeting minutes with MP supervisor

Our meeting minutes this week was more debugging with the supervisor. He speculated that the issue lies within finding the /bin directory in the NuttX shell. He tasked us to research and dig up more information and find where the NuttX version of the /bin directory is. From there, we may be able to grasp more understanding about the different commands available in the shell. He also suggested to email the pusher of the GitHub source codes to clarify about the /bin directory issue as well.

Weekly Self-Reflection

I am still very stressed about our current progress, as we have not been progressing much ever since getting the NuttX shell. There are not enough resources and documentations to help us in this project and we all have been trying our best to help each other and tirelessly scouring for resources and trying to understand how things work. I have also been having sleepless nights trying to find solutions and trying out different ways to compile source codes and trying to edit configuration files in hopes of seeing something new every time, but to no avail. Despite constantly meeting dead-ends, I still believe that as a group, with more hard work and effort together, we can hopefully progress in the following weeks to come.

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Student Name: Isaac Tan	**Adm No:** 2100902B
Supervisor Name: Yap Chern Nam	**Week No:** 05

Tasks Completed

Not much progress since the previous week. We are still trying to make the LED lights blink, as well as the MQTT part to work. The LED lights sort of work, but it is not automated. We have gotten 2 scenarios; after getting in the NuttX shell, we have to manually enter commands in the shell to make it blink one at a time, but it is not automated blinking. The other scenario is after using the `picocom` command, there will be no shell shown, but there will be blinking lights.

Issue/Risk Tracking

We do not have enough resources, documentations, and information to we can make the LED lights blink. What we have now is the best outcome we have gotten, considering the lack of resources to produce this result. The MQTT however, still needs a lot of research and trying to understand how the source code works when the configurations are compiled together.

Meeting minutes with MP supervisor

During out meeting this week, we presented our progress so far with the LED lights and our struggles dealing with the MQTT. Our supervisor did not say much and just advised us to continue what we are doing. He does not expect much from us every week as he is very understanding of the extreme struggles we have been encountering due to the lack of resources to work on this project. He also understands that ESP32-S3 is very new in the industry and because of that, the documentations and resources for it are very limited to aid us in working with this project.

Weekly Self-Reflection

It is extremely stressful to continuously encounter dead ends and not knowing how to make things work due to the lack of resources. It is incredibly frustrating working on this very new project and I still experience sleepless nights trying stay up thinking how to make the MQTT work, but to no avail. My groupmates are also as hardworking as they still try to figure out how to work with the LED lights and helping me with MQTT research. It calms me down better knowing that I have teammates who are experiencing the same difficulties and struggles as me and I am not alone.

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Student Name: Isaac Tan	**Adm No:** 2100902B
Supervisor Name: Yap Chern Nam	**Week No:** 06

Tasks Completed

We managed to send a file through MQTT from the publisher to the subscriber through the MQTT broker by reconfiguring the MQTT-C file after compiling all the configurations in the `make menuconfig` page, and then running a `gcc` command to run the simple_subscriber file.

Issue/Risk Tracking

For secure boot using MCUboot, we suspect that it is the same issues as the rest of the other parts of the project, which is the menuconfig page. There are too many dependencies in there that we do not know of and there are no resources or enough information to tell us exactly which options to use. Turning on one option may make another option appear in another section that we are unaware of.

Meeting minutes with MP supervisor

During our meeting minutes this week, we drew out an elaborate recap on the whiteboard of what we have accomplished so far, and listed the numerous struggles we have encountered that seem to be unsolvable due to the lack of resources and documentations online. We drew and gave a detailed explanation to our supervisor and he suggested to stop working on the MQTT, and continue with the MCUboot bootloader into the device, and see if there will be any progress with it. He also suggested some ways on how to present our MQTT findings during the final presentation and what to say, since we have not been able to make it work for weeks. He mentioned that we can ignore the security aspects of this project since we can barely make the fundamental parts work. He is very understanding of the difficulty of this project and we all appreciate his leniency towards us.

Weekly Self-Reflection

After seeing the recap drawing on the whiteboard, it was very demoralising seeing how much effort we have all put in only to see parts of the project working at the bare minimum. Even though they work at the bare minimum, we are still unable to integrate them to the general project. So, we only have parts of the project that work by themselves and not in the overall project. Despite being demoralised, I am still extremely grateful for my supervisor at how understanding he is towards us and this project.

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Project Title: OT Security Patch Management System	
Student Name: Isaac Tan	**Adm No:** 2100902B
Supervisor Name: Yap Chern Nam	**Week No:** 07

Tasks Completed

Same as previous weeks, we are still trying to perform secure boot using MCUboot, but nothing seems to be working. We are still constantly trying, hence there are no tasks completed this week. We also do not have any new findings this week to update each other about.

Issue/Risk Tracking

We still have no clue what the issue is, but our main suspicion is still the menuconfig page. As said previously as well, there are too many options for us to know which relates to each other. Furthermore, there also no documentations to tell us which options are related. Tutorials online are also useless, as they are mostly for previous versions of ESP32, like ESP32-S1 or ESP32-S2. As the versions are all very different from each other, it is impossible to follow exactly what the tutorial says. The menuconfig pages are completely different and there are options that are in S1 that are not in S3.

Meeting minutes with MP supervisor

Our supervisor failed to show up for this week's meeting despite constant persuasion and did not give a reason or update us beforehand. We also did not update him on our progress for this week.

Weekly Self-Reflection

As the deadline approaches, we all decided to stop working on the project and start discussing on the presentation and pitching slides. If we cannot do anything about the MCUboot and menuconfig, then we have no choice but to raise this up during the presentation and explain to the examiner that the lack of resources and documentation for this project has been extremely inefficient and there are many parts of this project that cannot be done. I felt very hopeless knowing that we have to stop the project already, given that all the weeks and hours of effort and attempts have been leading us nowhere. However, I am still very grateful for my teammates who also persevered on even when they probably have met more dead ends than me, and they still did not stop trying.

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Project Title: OT Security Patch Management System	
Student Name: Isaac Tan	**Adm No:** 2100902B
Supervisor Name: Yap Chern Nam	**Week No:** 08

Tasks Completed We are still trying to perform secure boot using MCUboot, but there seems to be no progress for the past few weeks. Hence, we have no tasks completed this week.
Issue/Risk Tracking As said previously, we do not know what exactly is the issue. We can only assume it is the dependencies in the configuration page, and we do not know which exact configuration options to turn on or off. Either that, or it could be a hardware issue this whole time. As the resources we have online have proven inefficient, we have no clue how to properly perform secure boot using MCUboot.
Meeting minutes with MP supervisor During our meeting minutes this week, our supervisor finally showed up to meet us after 2 weeks of constant pestering. He comforted us that though our project is not fully working well, at least we have a NuttX shell, Wi-Fi, MQTT, and LED Lights somewhat working that can be presented. I am extremely grateful for his understanding towards this project and did not ask us for much, neither did he complain or nag about how or project have nothing much working as he envisioned. We also further discussed about the things to say during presentation and pitching.
Weekly Self-Reflection After this week's meeting with the supervisor, it calmed me down even more as he told us not to worry so much and just present the things we have. It was a very stressful past 8 weeks of constantly working on this project to always meet dead-ends and not having all the parts completely working well, but it was assuring after my supervisor mentioned that previous projects took 3 MP batches to complete. So, it does not matter if our project is fully working or not. What matters is we all poured all our best efforts into this project and we have at least a few findings and parts to present.

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Student Name: Isaac Tan	**Adm No:** 2100902B
Supervisor Name: Yap Chern Nam	**Week No:** 09

Tasks Completed
No technical tasks relating to the project were completed. We stopped working on the project and started prioritising our time this week into making the presentation slides and preparing for our final presentation. We also did further modify our final documentation.

Issue/Risk Tracking
No issues were faced this week as we did no further work in the project.

Meeting minutes with MP supervisor
During our meeting minutes this week, we did a final run through of a mock presentation to our supervisor and he gave us his useful feedback, like what and what not to say that would cost us our presentation marks. He understands our frustration in this project and even advised us to emphasise more on presenting the pain that we went through during our final presentation.

Weekly Self-Reflection
As the project nears the end, I have not much thoughts left on this project as I really wanted this to end. I am happy that we all finally stopped working on the project and just starting to prepare the presentation slides to finally give our findings to our examiner and just emphasise how much pain and suffering we have gone through to get this little results.

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Student Name: Isaac Tan	**Adm No:** 2100902B
Supervisor Name: Yap Chern Nam	**Week No:** 10

Tasks Completed
Continued working on the presentation slides and script, final reflection report, and last changes to the final technical documentation.

Issue/Risk Tracking
No issues faced this week as we were all working on our presentation slides and personal reflection reports.

Meeting minutes with MP supervisor
During our meeting minutes this week, we did another final run through of a mock presentation to see what else can be changed and improved on. Our supervisor gave us good feedback throughout the mock presentation and was very nice towards us despite not having much results throughout this entire project. He further advised us on the specific things to say that can potentially gain a few presentation marks, and the things not to say that would cost us the marks. He has been a very good supervisor to us and we greatly appreciate his presence and advise for us throughout this entire project.

Weekly Self-Reflection
This is the final week of major project and we are all anticipating to get the presentation over and done with. I am extremely tired and exhausted and I have been needing a break for a very long time. We have all put in our best efforts into this project and despite the lack of results, I am confident that we will do well in our presentation by showing our results and findings for future Major project batches to work on, and to convince our examiner of the lack of resources that we have been going through in these 10 weeks.